

CSE 321 Homework 6

Fall 2015 Gebze Technical University

1. You are given an array of jobs where every job has a deadline and an associated profit if the job is finished before the deadline. Every job takes a single unit of time; i.e., the minimum possible deadline for any job is 1.

Considering that only one job can be scheduled at a time, the problem is how to maximize the total profit if only one job can be scheduled at a time.

Examples:

Input: Four Jobs with the following deadlines and profits

JobID	Deadline	Profit
a	4	20
b	1	10
c	1	40
d	1	30

Output: The following is a sequence of jobs yielding maximum profit
c, a

Input: Five Jobs with the following deadlines and profits

JobID	Deadline	Profit
a	2	100
b	1	19
c	2	27
d	1	25
e	3	15

Output: The following is a sequence of jobs yielding maximum profit
c, a, e

- a) Propose a greedy algorithm to address this problem.
 - b) Analyze the complexity of your algorithm
 - c) Does your greedy algorithm find the optimal solution? Why or why not?
 - d) Implement your algorithm.
-
2. Given n cities and distances between every pair of cities, the problem is to select k cities to place warehouses such that the maximum distance of a city to a warehouse is minimized.
 - a) Propose a greedy algorithm to address this problem.

- b) Analyze the complexity of your algorithm
 - c) Does your greedy algorithm find the optimal solution? Why or why not?
 - d) Implement your algorithm.
3. You are given n activities with their start and finish times. The problem is to select the maximum number of activities that can be performed by a single person, assuming that a person can only work on a single activity at a time.

Example:

Consider the following 6 activities.

`start[] = {1, 3, 0, 5, 8, 5};`

`finish[] = {2, 4, 6, 7, 9, 9};`

The maximum set of activities that can be executed

by a single person is {0, 1, 3, 4}

- a) Propose a greedy algorithm to address this problem.
 - b) Analyze the complexity of your algorithm
 - c) Does your greedy algorithm find the optimal solution? Why or why not?
4. There are n ropes with different lengths. We need to connect these ropes into one rope. The cost of connecting two ropes is equal to sum of their lengths. We need to connect the ropes with minimum cost.
- For example, if we are given 4 ropes with lengths 4, 3, 2 and 6, we can connect them in following ways:

- First connect ropes of lengths 2 and 3. Now we have three ropes with lengths 4, 6 and 5.
- Now connect ropes of lengths 4 and 5. Now we have two ropes with lengths 6 and 9.
- Finally, connect the two ropes and all ropes have connected.

Total cost for connecting all ropes is $5 + 9 + 15 = 29$, which is the optimum one. Other ways of connecting ropes would have either the same or higher cost. For example, if we connect 4 and 6 first (we get three strings of 3, 2 and 10), then connect 10 and 3 (we get two strings of 13 and 2), and finally connect 13 and 2, then the total cost in this way is $10 + 13 + 15 = 38$.

- a) Propose a greedy algorithm to address this problem.
- b) Analyze the complexity of your algorithm
- c) Does your greedy algorithm find an optimal solution? Why or why not?